

# Bandwidth Selection and Multivariate KDE

Donghyun Kim

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## KDE Bandwidth Selection

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# Normal Scale bandwidth selector

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We know the asymptotic MISE of KDE

$$\frac{1}{4}h^4 \{\mu_2(K)\}^2 \int \{p''(x)\}^2 dx + n^{-1}h^{-1} \int K^2(x) dx + o(h^4 + n^{-1}h^{-1})$$

and sum of leading terms is minimized by

$$h_{opt} = \left[ \frac{\int K^2}{(\mu_2(K))^2 \int (p'')^2} \right]^{\frac{1}{5}} n^{-\frac{1}{5}}$$

$h_{NS}$  simply replace  $p(x)$  for  $\phi_{\hat{\sigma}}$ . Using  $\int (\phi_{\hat{\sigma}}'')^2 = \frac{3}{8\hat{\sigma}^5\sqrt{\pi}}$ , we get

$$\hat{h}_{NS} = \left[ \frac{8\sqrt{\pi} \int K^2}{3(\mu_2(K))^2} \right]^{\frac{1}{5}} n^{-\frac{1}{5}} \hat{\sigma}$$

# Oversmoothing

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(Terrell, 1990, JASA, 85, 470-477)

$$\sup_{p \in \mathcal{F}(\sigma)} \left( \frac{\int K^2}{\mu_2(K)^2 \int (p'')^2} \right) = \left( \frac{243 \int K^2}{35 \mu_2(K)^2} \right) \sigma$$

where  $\mathcal{F}(\sigma)$  denotes class of densities with variance being  $\sigma^2$

This suggests oversmoothing bandwidth:

$$\hat{h}_{OS} = \left( \frac{243 \int K^2}{35 \mu_2(K)^2} \right)^{1/5} \hat{\sigma} n^{-1/5}$$

# Proof

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Among densities  $f$  defined on the real line with specified variance  $\sigma^2$ , the member of scale family of

$$h(x) = \begin{cases} \frac{35}{32}(1 - x^2)^3, & |x| \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

minimizes  $\int f''(x)^2 dx$ .

proof. For arbitrary density function  $f(x)$  that have 0 mean and  $\sigma^2$  variance,  $e(x) = f(x) - h(x)$  has  $\int e(x)dx = \int x^2 e(x)dx = 0$  and  $e(x) \geq 0$  for  $|x| \geq 1$ .

$$\int f''(x)^2 dx = \int h''(x)^2 dx + \int e''(x)^2 dx + 2 \int h''(x)e''(x) dx,$$

$$\begin{aligned} \int h''(x)e''(x) dx = \\ \frac{105}{2} (e(1) + e(-1)) + \frac{315}{4} \int_{-1}^x (1 - 5x^2)e(x) dx + \frac{315}{4} \int_{|x|>1} (5x^2 - 1)e(x) dx \geq 0 \end{aligned}$$

# Least squares cross-validation

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- Motivation:

$$\text{MISE}(\hat{p}) = \mathbb{E} \int \hat{p}(x)^2 dx - 2\mathbb{E} \int \hat{p}(x)p(x)dx + \int p(x)^2 dx$$

$\int p(x)^2 dx$  is a constant

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$$E[\text{LSCV}(h)] = \text{MISE}(\hat{p}) - \int p(x)^2 dx$$

with

$$\text{LSCV}(h) = \int \hat{p}(x)^2 dx - \frac{2}{n} \sum_{i=1}^n \hat{p}_{-i}(X_i)$$

## LSCV details

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Define leave-one-out estimator  $\hat{p}_{-i}(x) = \frac{1}{n-1} \sum_{j \neq i} K_h(x - X_j)$ . Then,

$$\begin{aligned} E(\hat{p}_{-i}(X_i)) &= E(K_h(X_1 - X_2)) = E(E(K_h(X_1 - X_2) | X_2)) \\ &= E\left[\int K_h(x - X_2)p(x)dx\right] = E\int \hat{p}(x)p(x)dx \end{aligned}$$

If  $K$  is symmetric,

$$\begin{aligned} \text{LSCV}(h) &= \int \hat{p}(x)^2 dx - \frac{2}{n} \sum_{i=1}^n \hat{p}_{-i}(X_i) \\ &= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n (K_h * K_h)(X_i - X_j) - \frac{2}{n(n-1)} \left(\sum \sum\right)_{i \neq j} K_h(X_i - X_j) \\ \hat{h}_{\text{LSCV}} &= \operatorname{argmin}_{h>0} \text{LSCV}(h) \end{aligned}$$



# K-fold

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- Partition data into K subsets  $A_1, \dots, A_K$
- Define:

$$\text{LSCV}_{K\text{-fold}}(h) = \int \hat{p}(x)^2 dx - \frac{2}{K} \sum_{l=1}^K \left( \frac{1}{|A_l|} \sum_{i \in A_l} \hat{p}_{-A_l}(X_i) \right)$$

- $\hat{h}_{K\text{-fold LSCV}} = \arg \min_{h>0} \text{LSCV}_{K\text{-fold}}(h)$

Approximation of  $\text{MISE}(h)$ :

$$\text{MISE}(h) \approx \frac{1}{nh} \int K^2 + \frac{1}{4} h^4 \mu_2(K)^2 \int (p'')^2$$

Expected squared norm of  $f''$  estimator:

$$\begin{aligned} \mathbb{E} \left[ \int (\hat{p}''(x; h))^2 dx \right] &= \int \text{Var}(\hat{p}''(x)) + \int (\mathbb{E} \hat{p}''(x))^2 \\ &= (n^{-1} h^{-5} \int (K'')^2 + o(n^{-1} h^{-5})) + \int (p''(x) + \frac{1}{2} \mu_2(K) p^{(4)}(x) h^2 + o(h^2))^2 dx \\ &= \int (p'')^2 + n^{-1} h^{-5} \int (K'')^2 + O(h^2) \end{aligned}$$

where  $K$  is of 2nd order.

# Biased Cross Validation

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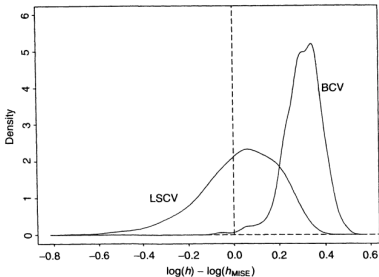
$$\hat{h}_{\text{BCV}} = \arg \min_{h>0} \text{BCV}(h)$$

$$\begin{aligned} \text{BCV}(h) &= \frac{1}{nh} \int K^2 + \frac{1}{4} h^4 \mu_2(K)^2 \left[ \int (\hat{p}''(x; h))^2 dx - n^{-1} h^{-5} \int (K'')^2 \right] \\ &= \frac{1}{nh} \int K^2 + \frac{1}{4} h^4 \mu_2(K)^2 \left[ n^{-2} h^{-4} \sum_{i=1}^n \sum_{j=1}^n (K'' * K'')_h(X_i - X_j) \right. \\ &\quad \left. - n^{-1} h^{-5} (K'' * K'')(0) \right] \\ &= \frac{1}{nh} \int K^2 + \frac{1}{4} h^4 \mu_2(K)^2 n^{-2} h^{-4} \sum_{i \neq j} (K'' * K'')_h(X_i - X_j) \end{aligned}$$

Use bandwidth  $h$  to estimate  $\int (p'')^2$

# BCV

- $BCV(h)$  may have multiple local minima too.
- $n^{1/10} \left( \frac{\hat{h}_{BCV}}{h_{opt}} - 1 \right) \xrightarrow{d} N(0, \sigma_{BCV}^2)$
- Typically,  $\sigma_{BCV}^2 < \sigma_{LSCV}^2$ .



- To improve  
Expand MISE to higher order terms + higher order kernel(bias of KDE)

# SE Rule Idea

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$$h = \left[ \frac{\int K^2}{\mu_2(K)^2 \tilde{\theta}_{2,0}(h)} \right]^{1/5} n^{-1/5}$$

$$\tilde{\theta}_{m,0}(h) = \hat{\theta}_m(h) - n^{-2} h^{-2m-1} \int (K^{(m)})^2$$

$$= n^{-2} h^{-2m} (-1)^m \left( \sum \sum \right)_{i \neq j} ((K^{(m)} * K^{(m)})_h (X_i - X_j))$$

where  $\hat{\theta}_m(h) = \int (\hat{p}^{(m)}(x; h))^2 dx$ . To estimate  $\int (p'')^2$ , use

$$\mathbb{E}[\hat{\theta}_m(h)] = \mathbb{E} \left[ \int (\hat{p}^{(m)}(x; h))^2 dx \right] = \int (p^{(m)})^2 + n^{-1} h^{-2m-1} \int (K^{(m)})^2 + O(h^2)$$

# SE rule idea

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- Use bandwidth  $g(h)$ , kernel  $L$  to estimate  $\int (p'')^2$

## Multivariate KDE

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# d-variate kernel

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Let  $\mathbf{X}_1, \dots, \mathbf{X}_n$  be iid d-variate random vectors with a density  $p : \mathbb{R}^d \rightarrow [0, \infty)$ . d-variate kernel function  $\kappa : \mathbb{R}^d \rightarrow \mathbb{R}$  s.t.  $\int \kappa(\mathbf{x}) d\mathbf{x} = 1$

- Product kernel

$$\kappa(\mathbf{x}) = \prod_{j=1}^d K(x_j)$$

- Radially symmetric kernel

$$\kappa(\mathbf{x}) = K(\mathbf{x}^T \mathbf{x})$$



# Bandwidth types

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- Single bandwidth:

$$\hat{p}(\mathbf{x}) = \frac{1}{nh^d} \sum_{i=1}^n \kappa\left(\frac{\mathbf{x} - \mathbf{X}_i}{h}\right)$$

- Coordinate-wise bandwidth:

$$\hat{p}(\mathbf{x}) = \frac{1}{n \prod_{j=1}^d h_j} \sum_{i=1}^n \kappa\left(\frac{x_1 - X_{i1}}{h_1}, \dots, \frac{x_d - X_{id}}{h_d}\right)$$

- Bandwidth matrix:

$$\hat{p}(\mathbf{x}) = \frac{1}{n} \text{Det}(H)^{-1/2} \sum_{i=1}^n \kappa(H^{-1/2}(\mathbf{x} - \mathbf{X}_i))$$

with symmetric positive definite  $H$



# Asymptotic bias

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$$\left\{ \begin{array}{l} \text{Every entry of } D^2p = \left( \frac{\partial^2}{\partial x_i \partial x_j} p \right) \text{ is continuous at } x \text{ and is integrable} \\ |t|^3 \kappa(t) \rightarrow 0 \text{ as } |t| \rightarrow \infty, \\ \int |t|^2 \kappa(t) dt < \infty, \\ \int \kappa(t) dt = 1, \\ \int t \kappa(t) dt = 0, \\ \int t t^T \kappa(t) dt = \mu_2(\kappa) I \end{array} \right.$$

$$\Rightarrow \text{bias}(\hat{p}(x; H)) = \frac{1}{2} \mu_2(\kappa) \text{tr}(D^2p(x)H) + o(\text{tr}(H))$$

# Asymptotic bias

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Using multivariate Taylor expansion

$$p(x + y) = p(x) + y^T D_p(x) + \frac{1}{2} y^T \mathcal{H}_p(x) y + o(y^T y)$$

in case of all entries of Hessian matrix are continuous in a neighborhood of  $x$ ,

$$\begin{aligned} E[\hat{p}(x)] &= \int K_H(x - u) p(u) du = \int K(z) p(x - H^{1/2} z) dz \\ &= \int K(z) \{ p(x) - (H^{1/2} z)^T D_p(x) + \frac{1}{2} (H^{1/2} z)^T \mathcal{H}_p(x) H^{1/2} z \} dz + o(\text{tr}(H)) \\ &= p(x) + \frac{1}{2} \mu_2(K) \text{tr}(H \mathcal{H}_p(x)) + o(\text{tr}(H)) \end{aligned}$$

# Asymptotic variance

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$$\begin{cases} |t|K^2(t) \rightarrow 0 \text{ as } |t| \rightarrow \infty, \\ \int K^2(t)dt < \infty \end{cases}$$

$$\Rightarrow \text{var}(\hat{p}(x; H)) = n^{-1} \text{Det}(H)^{-1/2} \int K^2(t)dt p(x) + o(n^{-1} \text{Det}(H)^{-1/2})$$

# Asymptotic variance

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$$\begin{aligned}\text{Var}\hat{p}(x; H) &= \frac{1}{n}|H|^{-1/2} \int K(z)^2 p(x - H^{1/2}z) dz \\ &\quad - \frac{1}{n} \left\{ \int K(z) p(x - H^{1/2}z) \right\}^2 \\ &= \frac{1}{n}|H|^{-1/2} \int (K(z))^2 dz p(x) + o(n^{-1}|H|^{-1/2})\end{aligned}$$

# Asymptotic MISE

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$$\int \text{bias}^2(\hat{p}(\mathbf{x}; \mathbf{H})) d\mathbf{x} = \frac{1}{4}\mu_2(K)^2 \int \text{tr}^2 \left( D^2 p(\mathbf{x}) \mathbf{H} \right) d\mathbf{x} + o \left( \text{tr}^2(\mathbf{H}) \right)$$

provided that each entry of  $D^2 p(\cdot)$  is continuous on  $\mathbb{R}^d$  and square integrable, that  $\int |\mathbf{t}|^2 K(\mathbf{t}) d\mathbf{t} < \infty$ ,  $\int K(\mathbf{t}) d\mathbf{t} = 1$ ,  $\int \mathbf{t} K(\mathbf{t}) d\mathbf{t} = 0$ , and  $\int \mathbf{t} \mathbf{t}^T K(\mathbf{t}) d\mathbf{t} = \mu_2(K) \mathbf{I}$ .

Similar to univariate case,

$$\begin{aligned} & \int \left\{ \int \left( \int_0^1 (1-w) z^\top H^{1/2} \left[ \nabla^2 f(x - w H^{1/2} z) - \nabla^2 f(x) \right] H^{1/2} z dw K(z) dz \right)^2 \right\} dx \\ &= \int \int \int \int_0^1 \int_0^1 (1-w_1)(1-w_2) \text{tr}(z_1 z_1^T K(z_1)) \text{tr}(z_2 z_2^T K(z_2)) [\nabla^2 f(x - w_1 H^{1/2} z_1) - \nabla^2 f(x)] \\ & \quad [\nabla^2 f(x - w_2 H^{1/2} z_2) - \nabla^2 f(x)] dw_1 dw_2 dz_1 dz_2 dx \end{aligned}$$

# Asymptotic MISE

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$$\int \text{var}(\hat{p}(\mathbf{x}; \mathbf{H})) d\mathbf{x} = n^{-1} \text{Det}(\mathbf{H})^{-1/2} \int K^2(\mathbf{t}) d\mathbf{t} + o\left(n^{-1} \text{Det}(\mathbf{H})^{-1/2}\right)$$

provided that  $p$  is continuous on  $\mathbb{R}^d$  and square integrable (or bounded), and that  $\int K^2(\mathbf{t}) d\mathbf{t} < \infty$ .

$$\begin{aligned} & \int \text{MSE}(\hat{f}(\mathbf{x}; \mathbf{H})) d\mathbf{x} \\ &= n^{-1} |\mathbf{H}|^{-1/2} \int K^2(\mathbf{t}) d\mathbf{t} + \frac{1}{4} \mu_2(K)^2 \int \text{tr}^2 \left( D^2 f(\mathbf{x}) \mathbf{H} \right) d\mathbf{x} + o(\text{tr}^2(\mathbf{H}) + n^{-1} |\mathbf{H}|^{-1/2}) \end{aligned}$$

## Using Product kernel, Bandwith vector

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$$\hat{p}(x) = \frac{1}{n} \sum_{i=1}^n \prod_{j=1}^d \left[ \frac{1}{h_j} K\left(\frac{x_j - X_{ij}}{h_j}\right) \right]$$

$$\mathbb{E}(\hat{p}(x)) - p(x) = \frac{1}{2} \mu_2(K) \text{tr}(H \mathcal{H}_p(x)) + o(\text{tr}(H))$$

$$= \frac{1}{2} \mu_2(K) \sum_{j=1}^d h_j^2 \frac{\partial^2}{\partial x_j^2} p(x) + o(h_1^2 + \dots + h_d^2).$$

$$\text{var}(\hat{p}(x)) = \frac{1}{n} |H|^{-1/2} \int K^2 p(x) + o(n^{-1} |H|^{-1/2})$$

$$= n^{-1} h_1^{-1} \dots h_d^{-1} \cdot \mu_0(K^2) p(x) + o\left(n^{-1} h_1^{-1} \dots h_d^{-1}\right).$$

Large  $nh_1 \dots h_d$  represent large number of observations contained in  $[x - h_1, x + h_1] \times \dots \times [x - h_d, x + h_d]$



# Optimal bandwidth

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$$\begin{aligned} \text{MISE}(h_1, \dots, h_d) &= \frac{1}{4} \mu_2(K)^2 \int \left( \sum_{j=1}^d h_j^2 \frac{\partial^2 p}{\partial x_j^2} \right)^2 dx \\ &\quad + \frac{1}{n \prod h_j} \mu_0(K^2) + o \left( \sum h_j^2 + \frac{1}{n \prod h_j} \right) \end{aligned}$$

If The sum of leading terms is minimized by  $h_{j,opt} = c_{j,opt} n^{-1/(d+4)}$  such that

$$\begin{aligned} &\int \left( \sum_{k=1}^d c_k^2 \frac{\partial^2}{\partial x_k^2} p(x) \right) \left( \frac{\partial^2}{\partial x_j^2} p(x) \right) dx \cdot c_j^3 \\ &= \frac{\mu_0(K^2)}{\mu_2(K)^2} \cdot \prod_{k \neq j} c_k^{-1}, \quad 1 \leq j \leq d. \end{aligned}$$

# Optimal Bandwidth

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Let  $h_j = c \cdot n^{-1/(d+4)}$  for all  $1 \leq j \leq d$ . In this case,

$$h_{j,opt} = c_{opt} \cdot n^{-1/(d+4)} \quad \text{with}$$

$$c_{opt} = \left( \frac{d \cdot \mu_0(K^2)}{(\mu_2(K))^2} \int \left( \sum_{j=1}^d \frac{\partial^2}{\partial x_j^2} p(x) \right)^2 dx \right)^{1/(d+4)}$$

# Bandwidth Selection

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## Marginal + Shrinkage:

1. Estimate marginal bandwidths  $\hat{h}_j^{\text{mg}}$  using LSCV
2. Grid search:  $\alpha \in G$ , minimize  $\text{LSCV}(\alpha \hat{h}_1^{\text{mg}}, \dots, \alpha \hat{h}_d^{\text{mg}})$

## Coordinate-wise Bandwidth Selection (CBS):

- Iteratively optimize each  $h_j$  fixing the others

# Curse of dimensionality

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- $\inf_{h_1, \dots, h_d > 0} \text{MISE}(\mathbf{h})$  decays as  $n^{-4/(d+4)}$
- Volume of effective neighborhood:  $\prod h_j \sim (0.1)^d$
- **Implication:** very few points contribute in high dimensions

# Freeing from the curse of dimensionality

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- Projection pursuit

$$p(\mathbf{x}) = p_0(\mathbf{x}) \prod_{j=1}^{d_1} f_j(\boldsymbol{\beta}_j^\top \mathbf{x}), \quad d_1 \leq d$$

- Multiplicative model

$$p(\mathbf{x}) = \prod_{j=1}^d f_j(x_j), \quad \mathbf{x} \in S \subset \mathbb{R}^d$$

# References

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- NonParametric Estimation-Lecturenote-2
- NonParametric Estimation-Lecturenote-3
- Kernel Smoothing